

SEWARD, AK, USA

WMO: 702770

Lat: 60.128N

Lon: 149.417W

Elev: 7

StdP: 101.24

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26438

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.4	-13.3	-23.8	0.4	-13.2	-22.0	0.5	-12.0	15.2	-10.6	14.1	-9.3	8.8	20	0.721

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.0	21.6	14.0	19.2	13.4	17.6	13.1	15.5	19.5	14.6	18.0	13.9	16.8	4.8	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.7	9.8	16.5	12.9	9.3	15.4	12.4	9.0	14.9	43.1	19.3	40.7	18.0	38.6	16.7	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	12.5	11.1	9.6	DB	-16.9	26.3	3.4	2.9	-19.4	28.4	-21.4	30.1	-23.3	31.7	-25.8	33.8
(5)				WB	-17.4	16.8	3.3	2.1	-19.8	18.3	-21.7	19.5	-23.6	20.7	-26.1	22.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.6	-3.1	-1.9	-0.5	3.5	7.8	11.2	13.3	13.2	9.8	4.7	-0.2	-2.4
	DBStd	6.99	5.57	4.71	3.99	2.68	2.85	2.60	2.33	2.20	2.25	2.98	4.23	4.55
	HDD10.0	2244	406	332	327	197	82	12	1	1	29	166	306	386
	HDD18.3	5001	664	565	586	446	327	214	158	161	256	424	556	644
	CDD10.0	291	0	0	0	1	13	49	103	100	24	1	0	0
	CDD18.3	5	0	0	0	0	0	1	3	1	0	0	0	0
	CDH23.3	33	0	0	0	0	1	7	18	8	0	0	0	0
	CDH26.7	4	0	0	0	0	0	1	2	1	0	0	0	0
Wind	WSAvg	3.8	4.9	4.4	4.5	3.5	3.3	2.9	2.6	2.8	3.2	3.9	4.8	5.0
Precipitation	PrecAvg	1660	160	138	101	97	109	63	57	135	251	269	159	187
	PrecMax	2448	646	338	312	247	405	224	153	359	662	610	641	500
	PrecMin	925	17	2	4	7	11	16	4	24	78	53	5	24
	PrecStd	389	148	89	73	73	85	42	37	88	135	146	143	143
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.3	6.6	9.3	15.2	21.0	23.6	26.1	23.9	18.8	12.8	8.9	7.1
		MCWB	4.1	3.8	5.1	7.7	11.2	14.8	16.3	15.5	12.0	8.1	6.4	5.1
	2%	DB	5.8	5.2	7.0	11.3	17.3	20.1	21.4	21.1	16.3	11.0	7.4	5.8
		MCWB	3.8	3.1	3.8	5.8	9.7	13.2	14.6	13.9	10.9	7.4	5.9	4.1
	5%	DB	4.5	4.1	5.5	9.2	14.7	17.4	18.9	18.8	14.6	9.9	6.3	4.5
		MCWB	3.0	2.4	2.9	5.3	9.0	12.6	14.2	13.4	10.7	7.2	5.0	3.2
	10%	DB	3.3	3.1	4.1	7.8	12.6	15.8	17.3	17.3	13.2	8.9	5.1	3.2
		MCWB	2.2	1.7	2.1	4.5	8.4	11.9	13.9	13.4	10.3	6.8	3.9	2.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.4	4.6	6.2	8.9	12.0	15.8	16.8	16.6	13.2	10.2	7.5	5.7
		MCDB	6.7	5.8	8.1	14.4	18.4	21.6	23.6	21.4	15.9	11.3	8.0	6.6
	2%	WB	4.1	3.5	4.7	6.8	10.6	14.1	15.6	15.4	12.4	8.9	6.3	4.3
		MCDB	5.3	4.8	6.3	9.8	15.2	18.1	19.9	19.0	14.5	9.7	7.1	5.4
	5%	WB	3.3	2.8	3.4	5.8	9.7	13.1	14.8	14.5	11.7	8.0	5.3	3.4
		MCDB	4.2	3.9	4.8	8.3	13.4	16.6	18.2	17.7	13.6	8.9	6.3	4.3
	10%	WB	2.4	2.1	2.6	5.1	8.8	12.2	14.1	13.8	11.1	7.3	4.1	2.4
		MCDB	3.2	3.1	3.8	7.4	12.1	15.0	16.9	16.4	12.8	8.4	5.0	3.2
Mean Daily Temperature Range	5% DB	MDBR	4.8	5.3	6.1	6.8	7.2	6.7	6.0	6.4	6.2	5.7	4.5	4.7
		MCDBR	5.3	5.6	7.1	9.7	11.4	11.1	10.6	10.4	8.5	6.8	5.1	5.4
		MCWBR	3.9	3.9	4.8	5.8	6.0	5.8	5.2	5.1	4.8	4.1	3.8	4.0
	5% WB	MCDBR	5.1	5.0	6.3	8.2	9.2	9.3	8.7	8.5	6.8	5.1	4.8	5.1
		MCWBR	3.8	3.7	4.6	5.4	5.3	5.2	4.8	4.9	4.3	3.7	3.8	4.0
Clear-Sky Solar Irradiance	taub	0.260	0.315	0.329	0.370	0.389	0.387	0.390	0.378	0.337	0.317	0.266	0.257	
	taud	2.081	2.097	2.108	2.047	2.100	2.252	2.303	2.338	2.434	2.269	2.070	1.942	
	Ebn at Noon	492	654	782	815	830	842	829	809	788	658	464	308	
	Edh at Noon	44	79	107	137	140	123	115	103	78	66	44	33	
All-Sky Solar Radiation	RadAvg	0.31	0.96	2.43	3.95	5.13	5.24	4.68	3.71	2.47	1.18	0.45	0.18	
	RadStd	0.04	0.17	0.31	0.45	0.43	0.28	0.43	0.43	0.37	0.13	0.08	0.02	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.42	+3.91	+3.66	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (26 neighbors)	+0.68	+2.15	+1.89	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page